PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Living through the psychological consequences of COVID-19 pandemic: a systematic review of effective mitigating interventions
AUTHORS	Lekagul, Angkana; Piancharoen, Peeraya; Chattong, Anamika; Suradom, Chawisa; Tangcharoensathien, Viroj

VERSION 1 – REVIEW

REVIEWER	Napoli, Christian Sapienza University of Rome, Medical surgical sciences and translational medicine
REVIEW RETURNED	27-Jan-2022

GENERAL COMMENTS	First of all, I would like to thank for the opportunity to review this paper. COVID-19 is an ongoing pandemic that has resulted in global health, economic and social crises. Moreover, addressing the psychological effects of infectious diseases outbreak is challenging, since most efforts and services are intended to protect physical wellbeing and less attention is paid to the psychological side. In this context, the paper under review is aimed at assessing interventions and their effectiveness in mitigating COVID-19 pandemic psychological consequences. The subject under study is certainly very important, especially in the historical period we are experiencing. The article presents interesting results but, but it is nevertheless believed that can be improved. I would like to encourage authors to consider some issues to be improved. Title: I suggest to improve, better specifying the objective of the study. Introduction: The authors do not frame their study within the vast body of literature that addressed the issue, after this it should make clearer what is the gap in the literature that is filled with this study. What is the international situation regarding the psychological effects of the pandemic? What is the possible international contribution of the study to the literature? What are the implications of the study? Discussion: I also suggest expanding what is the possible international contribution of the study to the literature, what are the implications of the study. ? The vaccination effects on psychological well-being is not investigate at all. Actually, the vaccination campaign is the first method to counteract the COVID-19 pandemic and sufficient vaccination coverage is conditioned by the people's acceptance of these vaccines. This issue must be discussed at least in the limits section (refer to articles with DOI: https://doi.org/10.3390/vaccines9111222).

REVIEWER	Roy, Deblina

	King George's Medical University, Department of Neuroscience, Section of Psychiatry
REVIEW RETURNED	29-Jan-2022

GENERAL COMMENTS	Dear Authors, This paper is well written and also addresses the need of the hour it is written to the point and has fairly simple language

REVIEWER	Sahin, Mustafa Ondokuz Mayis University
REVIEW RETURNED	08-Mar-2022

GENERAL COMMENTS	Dear Author, It is a well-written manuscript. I have some comments. In abstract section "For this systematic review....." you should write your study type. Specify the inclusion and exclusion criteria for the review. In method section Specify the date when each source was last searched or consulted. Best wishes

REVIEWER	tanveer, sandila University of Otago Christchurch, Department of Psychological Medicine
REVIEW RETURNED	16-Mar-2022

GENERAL COMMENTS	This is a worthy effort carried out on a prevailing issue that can inform practitioners in the clinical setting. The paper is submitted for publication in 2022, the authors must provide some justification on why the literature search is limited to studies published in 2020. Although, it would be ideal that a quick database search is carried out and the paper is updated accordingly. The authors must address the following issues. 1. The psychological impacts of the Covid-19 are well reported in the literature since 2020. Considering this, the introduction seems brief and lacks some solid information. 2. Although an extensive database search is carried out following standard guidelines but authors need to add more details for replicate efforts. 3. There are frequent repetitions in the results sections which need to be fixed. 4. The journal recommended number of Tables is 5 which exceeds in this paper. 5. In the discussion section, the ideas need to be organized in a way that the main findings are mentioned first followed by supplementary findings. This section needs major revision, as results could be presented in a more structured and insightful manner. 6. The significance of the study is missing before the limitations and the limitations and conclusion section need some work. To help address the above issues, several comments have been made throughout the paper. It is believed that this will enhance the quality of the

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Christian Napoli, Sapienza University of Rome

Comments to the Author:
First of all, I would like to thank for the opportunity to review this paper. COVID-19 is an ongoing pandemic that has resulted in global health, economic and social crises. Moreover, addressing the psychological effects of infectious diseases outbreak is challenging, since most efforts and services are intended to protect physical wellbeing and less attention is paid to the psychological side. In this context, the paper under review is aimed at assessing interventions and their effectiveness in mitigating COVID-19 pandemic psychological consequences.

The subject under study is certainly very important, especially in the historical period we are experiencing. The article presents interesting results but, but it is nevertheless believed that can be improved. I would like to encourage authors to consider some issues to be improved.

Title: I suggest to improve, better specifying the objective of the study.
RESPONSES
Similar to comment made by the Associate Editor; the title has been modified as followed.

"Living through the psychological consequences of COVID-19 pandemic: a systematic review of effective mitigating interventions"

The revised title reflects the objectives of the study, which is a systematic review of interventions which are effective in mitigating the psychological consequences from the COVID-19 pandemic.

Introduction: The authors do not frame their study within the vast body of literature that addressed the issue, after this it should make clearer what is the gap in the literature that is filled with this study. What is the international situation regarding the psychological effects of the pandemic? What is the possible international contribution of the study to the literature? What are the implications of the study?
RESPONSES
As suggested by the reviewer, in the Introduction section, we fully described the background literature.

Paragraph 1 provided background of COVID-19 and its impact. Paragraph 2 described about the direct and indirect impacts of COVID-19 pandemic on mental health, such as fear, feeling of hopelessness, despair, and grief. Paragraph 3 focuses on significant psychological impacts on certain population group such as health care workers from caring COVID patients, fear of transmitting disease to their family members. Paragraph 4 provides information from previous studies about interventions to mitigate psychological consequences in other public health emergencies such as MER-CoV, Ebola and current COVID-19 pandemic. Paragraph 5 mentioned the gaps of current interventions to mitigate psychological impacts from COVID-19 pandemic which requires better understanding through systematic review and if interventions are effective, to scale up in other countries as the pandemic has not yet concluded. Finally, the policy implication of this study at the last paragraph of introduction section. It says, "Findings from this review can inform health professionals and governments of effective interventions to prevent or mitigate mental health problems from the COVID-19 pandemic."

Discussion: I also suggest expanding what is the possible international contribution of the study to the literature, what are the implications of the study? The vaccination effects on psychological well-being is not investigate at all. Actually, the vaccination campaign is the first method to counteract the COVID-19 pandemic and sufficient vaccination coverage is conditioned by the people's acceptance of these vaccines. This issue must be discussed at least in the limits section (refer to articles with DOI: https://doi.org/10.3390/vaccines9111222).

RESPONSES

Thank you for pointing this out. The discussion section has been modified.

We renamed the section title to "Conclusion and policy implications".

We have added the suggested content in the last paragraph of the "conclusion and policy implications" and a suggested reference was included: https://doi.org/10.3390/vaccines9111222

"We recommend that these effective interventions can be replicated and applied in other country settings, for which modes of delivery can be adjusted in line with country resources, epidemic situation, and social and health systems contexts. In the first quarter of 2022, the pandemic is still far from over due to low immunization coverage in African countries and continued emergence of Variants of Concerns in the context of pressure to resuming the economic activities and opening the country borders. [REF: BMJ 2022;376:o311 http://dx.doi.org/10.1136/bmj.o311]. In ending the acute phase of pandemic, rapid scale up and promote vaccine acceptance are key policy actions. [REF: Gallè, F.; Sabella, E.A.; Roma, P.; Da Molin, G.; Diella, G.; Montagna, M.T.; Ferracuti, S.; Liguori, G.; Orsi, G.B.; Napoli, C. Acceptance of COVID-19 Vaccination in the Elderly: A Cross-Sectional Study in Southern Italy. Vaccines 2021, 9, 1222. https://doi.org/10.3390/vaccines9111222]"

Reviewer: 2
Dr. Deblina Roy, King George's Medical University, AIIMS

Comments to the Author:
Dear Authors,
This paper is well written and also addresses the need of the hour; it is written to the point and has fairly simple language

RESPONSES

Thank you for the comments.

Reviewer: 3
Dr. Mustafa Sahin, Ondokuz Mayis University

Comments to the Author:
Dear Author,

It is a well-written manuscript.
I have some comments.

In abstract section
"For this systematic review....." you should write your study type.
Specify the inclusion and exclusion criteria for the review.

RESPONSES

Many thanks for the suggestions; the search term, inclusion and exclusion criteria are fully addressed in Table 1 in the PICO framework.

In method section
Specify the date when each source was last searched or consulted.
RESPONSES
Agree. We added specific duration for searching “between January 2020 and September 2021” in the abstract and in method section.

Reviewer: 4
Dr. sandila tanveer, University of Otago Christchurch

*** Please find additional comments from this reviewer in the attached file ***
Comments to the Author:

This is a worthy effort carried out on a prevailing issue that can inform practitioners in the clinical setting. The paper is submitted for publication in 2022, the authors must provide some justification on why the literature search is limited to studies published in 2020. Although, it would be ideal that a quick database search is carried out and the paper is updated accordingly.
RESPONSES
Thank you very much for agreeing with us to the intention of this manuscript.

We appreciate the insightful comments and suggestions. We have read and fully responded to all the comments.

Publication included in the review between January 2020 and September 2021. We covered literature published from 2020 onwards as the COVID-19 pandemic emerged in early 2020.

The authors must address the following issues.
1. The psychological impacts of the Covid-19 are well reported in the literature since 2020. Considering this, the introduction seems brief and lacks some solid information.
RESPONSES
Thank you for pointing this out.

Similar to comment by reviewer 1, in the Introduction section, we fully described the background literature.

Paragraph 1 provided background of COVID-19 and its impact. Paragraph 2 described about the direct and indirect impacts of COVID-19 pandemic on mental health, such as fear, feeling of hopelessness, despair, and grief. Paragraph 3 focuses on significant psychological impacts on certain population group such as health care workers from caring COVID patients, fear of transmitting disease to their family members. Paragraph 4 provides information from previous studies about interventions to mitigate psychological consequences in other public health emergencies such as MER-CoV, Ebola and current COVID-19 pandemic. Paragraph 5 mentioned the gaps of current interventions to mitigate psychological impacts from COVID-19 pandemic which requires better understanding through systematic review to scale up effective interventions in other countries. Finally, the policy implication of this study at the last paragraph of introduction section.

2. Although an extensive database search is carried out following standard guidelines but authors need to add more details for replicate efforts.
RESPONSES

Thank you very much for your suggestion. In order to facilitate future similar studies, the search terms have been fully described in the methodology section.

3. There are frequent repetitions in the results sections which need to be fixed.

RESPONSES

We thank the reviewer for this comment. We have rearranged the results section. We have modified the tables according to your detailed comments in the PDF manuscript. In the revised manuscript, we keep five tables.

- We have deleted two tables:

1. Table 3 described tools for the assessment of psychological consequences (in Annex 2)
2. Table 6 summarise of outcomes of interventions and we have incorporated key contents in table 6 into table 3 and 4.

- According to the comment made in the PDF manuscript, we cannot merge table 3 and 4 due to the journal format does not allow one table extended to more than 2 pages. Threfore, the table 3 provided information of RCT and table 4 for non-RCT studies.

4. The journal recommended number of Tables is 5 which exceeds in this paper.

RESPONSES

Many thanks for your kind reminder. We had responded in comment 3 above, we kept only five tables in the revised manuscript.

5. In the discussion section, the ideas need to be organized in a way that the main findings are mentioned first followed by supplementary findings. This section needs major revision, as results could be presented in a more structured and insightful manner.

RESPONSES

Thank you very much for the comment.

We had re-structured the result into four logical sub-sections to respond to the three review questions

First, what interventions are used to mitigate psychological consequences in the population and in health care workers?

Second, what tools are used for assessing intervention outcome?

Third, what is the effectiveness of different interventions?

In the discussion section we have rearranged as follows:

- Study characteristics (it also includes discussion of the tools used for the assessment of psychological consequences)
- Interventions, measurement, and effectiveness
- Risk of biases (added according to the detailed comment in the attached file).
- Limitations of the study

6. The significance of the study is missing before the limitations and the limitations and conclusion section need some work.

RESPONSES

We agree with your comments. The limitation and conclusion sections have been updated.

In the discussion section, we have added limitations such as small sample sizes, post-intervention outcomes measurement.

To help address the above issues, several comments have been made throughout the paper. It is believed that this will enhance the quality of the paper and would enable it to make a meaningful contribution to the literature.
We appreciate the thorough reading and specific suggestions in the manuscript.
We have incorporated changes to reflect most of the suggestions provided by the reviewer. By this, the revised manuscript has been significantly improved.

VERSION 2 – REVIEW

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

REVIEWER	
REVIEW RETURNED	

GENERAL COMMENTS	

VERSION 2 – AUTHOR RESPONSE

VERSION 3 – REVIEW

REVIEWER	Napoli, Christian Sapienza University of Rome, Medical surgical sciences and translational medicine
REVIEW RETURNED	30-May-2022

GENERAL COMMENTS	the authors declare that a sentence was added "In ending the acute phase of pandemic, rapid scale up and promote vaccine acceptance are key policy actions. [REF:..... ". But I cannot see this sentence in the new vesion. Please add with the proper reference

REVIEWER	tanveer, sandila

	University of Otago Christchurch, Department of Psychological Medicine
REVIEW RETURNED	01-Jun-2022

GENERAL COMMENTS	I am quite delighted with the revision, and I want to commend the authors for all of their efforts in making this substantial contribution. The introduction, methods, and discussion sections are well laid-out. Overall, the paper provides succinct answers to research questions. The paper and tables still have some formatting issues. Otherwise, the paper is complete and ready for publication. I'd want to take this opportunity to congratulate the authors for this worthy effort and wish them well in their future endeavors.

VERSION 3 – AUTHOR RESPONSE

Reviewer: 1

Dr. Christian Napoli, Sapienza University of Rome

Comments to the Author:

the authors declare that a sentence was added "In ending the acute phase of pandemic, rapid scale up and promote vaccine acceptance are key policy actions. [REF:..... ". But I cannot see this sentence in the new version. Please add with the proper reference

RESPONSES

We apologise with this mistake.
In this version, we have added two references in third paragraph of section on conclusion and policy implication.

Reviewer: 4

Dr. sandila tanveer, University of Otago Christchurch

Comments to the Author:
I am quite delighted with the revision, and I want to commend the authors for all of their efforts in making this substantial contribution. The introduction, methods, and discussion sections are well laid-out. Overall, the paper provides succinct answers to research questions. The paper and tables still have some formatting issues. Otherwise, the paper is complete and ready for publication. I'd want to take this opportunity to congratulate the authors for this worthy effort and wish them well in their future endeavors.

RESPONSES

Thank you very much. We are grateful with
your valuable comments and constructive suggestions. Your insights help us to strengthen the manuscript.